

Zad 1.

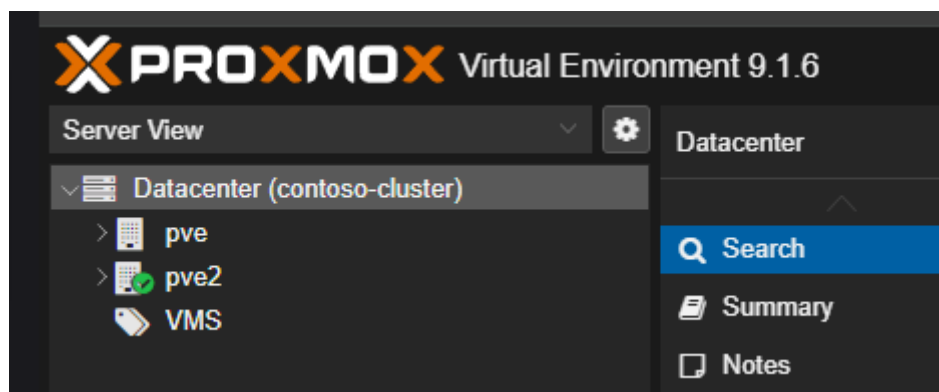
Celem zadania było utworzenie klastra z dwóch węzłów Proxmox VE oraz zapewnienie ich komunikacji w sieci wewnętrznej.

```
GNU nano 8.4 /etc/hosts *
127.0.0.1 localhost.localdomain localhost
192.168.80.128 pve1.contoso.com pve1
192.168.80.136 pve2.contoso.com pve2
# The following lines are desirable for IPv6 capable hosts
```

```
^C--- 192.168.80.128 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.169/0.202/0.235/0.033 ms
root@MD1:~# ping pve1
PING pve1.contoso.com (192.168.80.128) 56(84) bytes of data.
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=1 ttl=64 time=0.364 ms
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=2 ttl=64 time=0.179 ms
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=3 ttl=64 time=0.174 ms
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=4 ttl=64 time=0.213 ms
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=5 ttl=64 time=0.206 ms
64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=6 ttl=64 time=0.187 ms
^X64 bytes from pve1.contoso.com (192.168.80.128): icmp_seq=7 ttl=64 time=0.196
ms
^^
```

```
root@pve:~# pvecm create contoso-cluster --link0 192.168.1.10
Corosync Cluster Engine Authentication key generator.
Gathering 2048 bits for key from /dev/urandom.
Writing corosync key to /etc/corosync/authkey.
Writing corosync config to /etc/pve/corosync.conf
Restart corosync and cluster filesystem
root@pve:~# _
```

```
root@pve2:~# pvecm add pve.contoso.com --use_ssh
The authenticity of host 'pve.contoso.com (192.168.1.10)' can't be established.
ED25519 key fingerprint is SHA256:Ij5d6uslX2HP+vC0oumRbv7wSpN40spCtWuq+E6dh50.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
root@pve.contoso.com's password:
No cluster network links passed explicitly, fallback to local node IP '192.168.1.11'
copy corosync auth key
stopping pve-cluster service
backup old database to '/var/lib/pve-cluster/backup/config-1774177376.sql.gz'
waiting for quorum...OK
(re)generate node files
generate new node certificate
merge authorized SSH keys
generated new node certificate, restart pveproxy and pvedaemon services
successfully added node 'pve2' to cluster.
root@pve2:~#
```



Zad 2.

Celem zadania było skonfigurowanie mechanizmu quorum w klastrze Proxmox VE z wykorzystaniem zewnętrznego serwera Debian GNU/Linux jako dysku świadka.

```
GNU nano 8.4 /etc/ssh/sshd_config
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

```
# pvecm qdevice setup 192.168.100.200 -f
```

```
Done
root@pve:~# pvecm status
Cluster information
-----
Name:                contoso-cluster
Config Version:      3
Transport:           knet
Secure auth:         on

Quorum information
-----
Date:                Sun Mar 22 12:59:11 2026
Quorum provider:     corosync_votequorum
Nodes:               2
Node ID:             0x00000001
Ring ID:             1.24
Quorate:             Yes

Votequorum information
-----
Expected votes:      3
Highest expected:    3
Total votes:         3
Quorum:              2
Flags:               Quorate Qdevice

Membership information
-----
Nodeid      Votes   Qdevice Name
0x00000001   1     A,V,NM 192.168.1.10 (local)
0x00000002   1     A,V,NM 192.168.1.11
0x00000000   1           Qdevice
root@pve:~# _
```

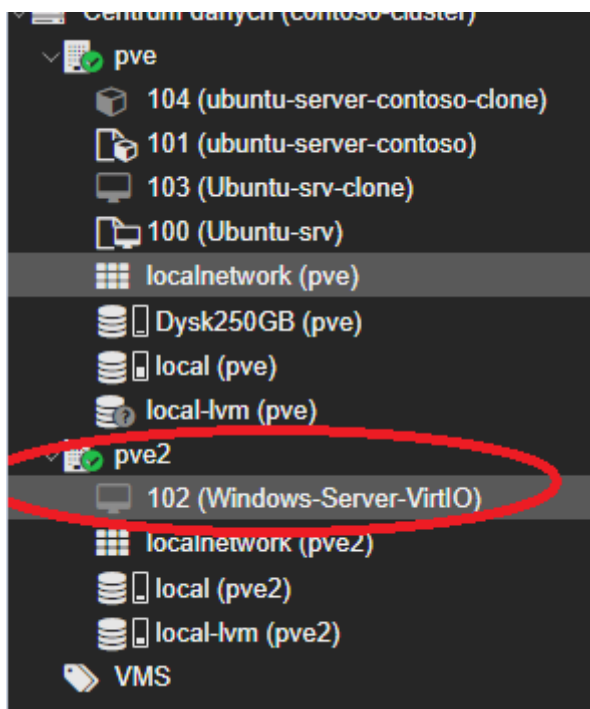
Zad 3.

Celem zadania było przeprowadzenie migracji maszyny wirtualnej między węzłami klastra Proxmox VE bez użycia współdzielonej pamięci.

Na pve -> pve2

```
root@pve:~# qm migrate 102 pve2 --targetstorage local-lvm --online
```

```
2026-03-22 13:17:04 volume 'Dysk250GB:vm-102-disk-2' is 'local-lvm:vm-102-disk-2' on the target
Logical volume "vm-102-disk-0" successfully removed.
Logical volume "vm-102-disk-1" successfully removed.
Logical volume "vm-102-disk-2" successfully removed.
2026-03-22 13:17:05 migration finished successfully (duration 00:04:25)
root@pve:~#
```



Zad 4.

Celem zadania było utworzenie i konfiguracja zasobu iSCSI na serwerze Debian GNU/Linux.

```
/> cd /backstores/fileio
/backstores/fileio> create disk1 /iscsi/disk1.img 30G
Created fileio disk1 with size 32212254720
/backstores/fileio> cd /iscsi
/iscsi> create iqn.2021-09.private:storage.targetmd1
Created target iqn.2021-09.private:storage.targetmd1.
Created TPG 1.
Global pref auto_add_default_portal=true
Created default portal listening on all IPs (0.0.0.0), port 3260.
/iscsi> cd /iscsi.iqn.2021-09.private:storage.targetmd1/tpg1/luns
No such path /iscsi.iqn.2021-09.private:storage.targetmd1
/iscsi> cd /iscsi/iqn.2021-09.private:storage.targetmd1/tpg1/luns
/iscsi/iqn.20...md1/tpg1/luns> create /backstores/fileio/disk1
Created LUN 0.
/iscsi/iqn.20...md1/tpg1/luns> cd /iscsi/iqn.2021-09.private:storage.targetmd1/tpg1/acls
/iscsi/iqn.20...md1/tpg1/acls> create iqn.2021-09.com.private:clientsrv1
Created Node ACL for iqn.2021-09.com.private:clientsrv1
Created mapped LUN 0.
/iscsi/iqn.20...md1/tpg1/acls>
```

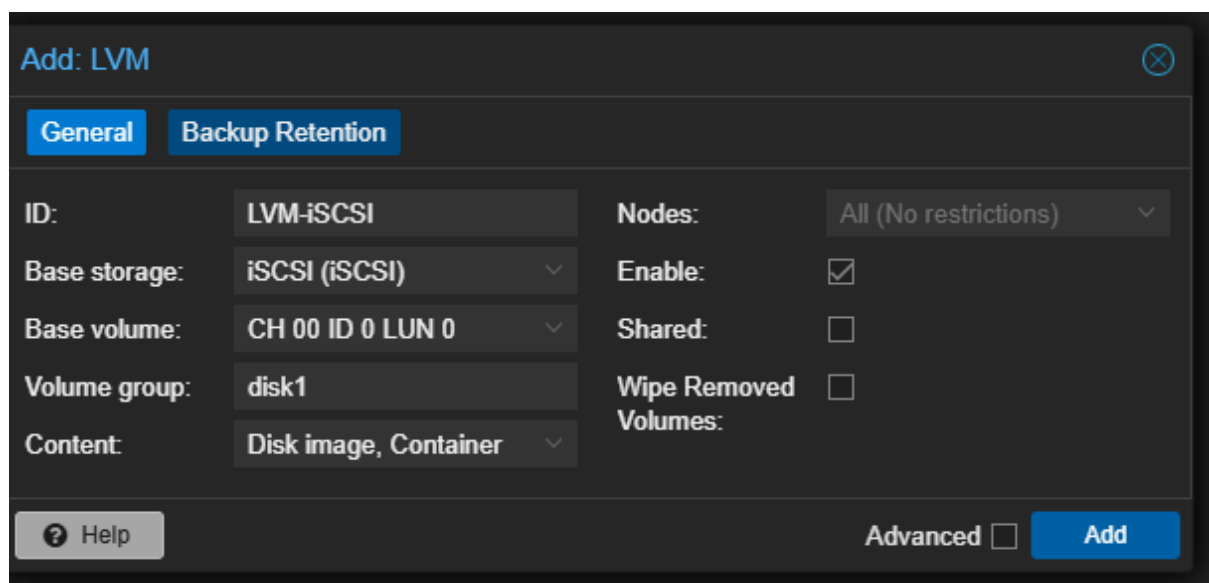
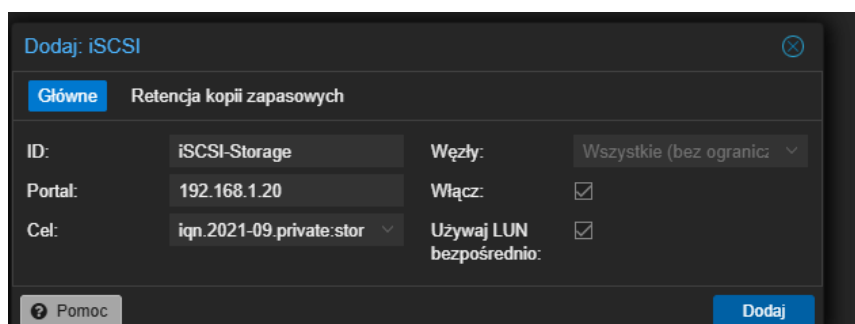
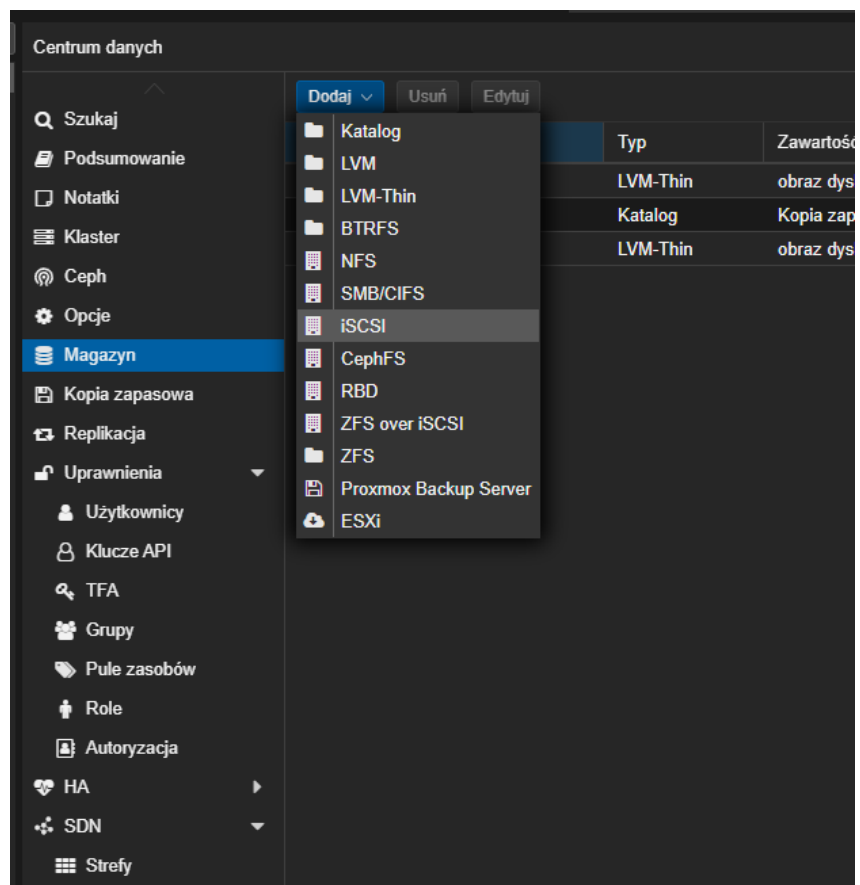
```
/iscsi/iqn.20...md1/tpg1/acls> cd /
/> saveconfig
Configuration saved to /etc/rtplib-fb-target/saveconfig.json
/>
```

```
Global pref auto_save_on_exit=true
Last 10 configs saved in /etc/rtplib-fb-target/backup/.
Configuration saved to /etc/rtplib-fb-target/saveconfig.json
root@MD1:~#
```

Zad 5.

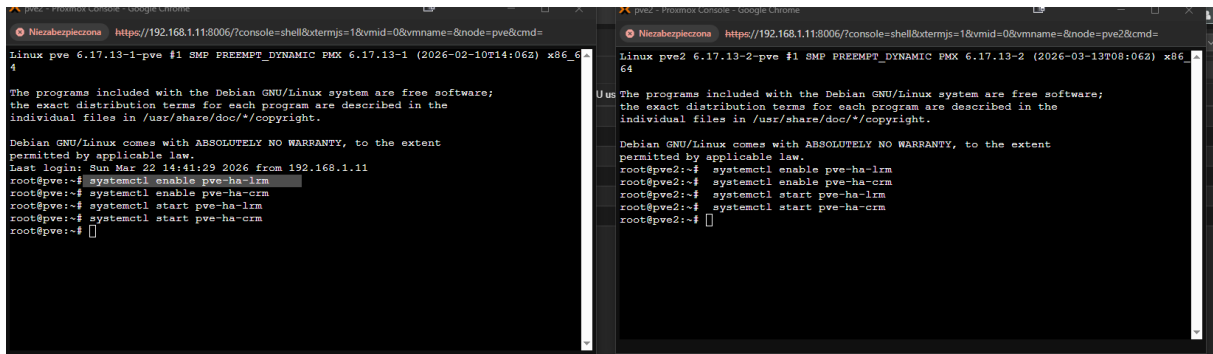
Celem zadania było podłączenie zasobu iSCSI do klastra Proxmox VE oraz przetestowanie migracji maszyny wirtualnej „na żywo”.

```
root@pve:~# iscsiadm -m node --login
iscsiadm: default: 1 session requested, but 1 already present.
Login to [iface: default, target: iqn.2021-09.private:storage.targetmd1, portal: 192.168.1.20,3260] successful.
iscsiadm: Could not log into all portals
root@pve:~# lsblk
NAME                                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0    50G  0 disk
|-sda1                              8:1    0 1007K  0 part
|-sda2                              8:2    0   512M  0 part
`-sda3                              8:3    0   49.5G  0 part
  |-pve-swap                        252:2    0    3.6G  0 lvm  [SWAP]
  |-pve-root                        252:3    0   45.9G  0 lvm  /
  sdb                                8:16    0   250G  0 disk
  |-Dysk250GB-Dysk250GB_tmeta      252:0    0    2.5G  0 lvm
  | `--Dysk250GB-Dysk250GB-tpool 252:4    0  244.9G  0 lvm
  |   `--Dysk250GB-Dysk250GB      252:5    0  244.9G  1 lvm
  |-Dysk250GB-Dysk250GB_tdata      252:1    0  244.9G  0 lvm
  |-Dysk250GB-Dysk250GB-tpool      252:4    0  244.9G  0 lvm
  | `--Dysk250GB-Dysk250GB      252:5    0  244.9G  1 lvm
  sdc                                8:32    0    30G  0 disk
  sdd                                8:48    0    30G  0 disk
  sr0                               11:0    1    1.7G  0 rom
```



Zad 6.

Celem zadania było włączenie wysokiej dostępności (HA) dla maszyny wirtualnej w klastrze Proxmox VE oraz sprawdzenie jej działania.



```
Linux pve 6.17.13-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.13-1 (2026-02-10T14:06Z) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Mar 22 14:41:29 2026 from 192.168.1.11
root@pve1:~# systemctl enable pve-ha-lrm
root@pve1:~# systemctl enable pve-ha-crm
root@pve1:~# systemctl start pve-ha-lrm
root@pve1:~# systemctl start pve-ha-crm
root@pve1:~#

Linux pve2 6.17.13-2-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.13-2 (2026-03-13T08:06Z) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@pve2:~# systemctl enable pve-ha-lrm
root@pve2:~# systemctl enable pve-ha-crm
root@pve2:~# systemctl start pve-ha-lrm
root@pve2:~# systemctl start pve-ha-crm
root@pve2:~#
```

```
root@pve2:~# qm list
      VMID NAME                STATUS      MEM(MB)   BOOTDISK(GB) PID
      ---  ---                -
      102 Windows-Server-VirtIO stopped      8192         64.00  0
root@pve2:~# ha-manager add vm:102
root@pve2:~#
```

```
root@pve2:~# ha-manager status
quorum OK
master pve2 (active, Sun Mar 29 16:12:47 2026)
lrm pve (idle, Sun Mar 29 16:12:50 2026)
lrm pve2 (active, Sun Mar 29 16:12:50 2026)
service vm:102 (pve2, starting)
root@pve2:~#
```